Hybrid Role-Aware Structural Modeling and Semantic Extraction for Long-Form Radio Broadcasts

Abstract

Long-form radio broadcasts consisting of DJ, music, and advertisements challenge conventional audio segmentation models. We propose a hybrid framework integrating speaker diarization, acoustic analysis, and audio fingerprint matching. Specifically, we identify DJ blocks using speaker diarization, separate music and advertisement blocks via acoustic features combined with duration and speaker-dynamics priors, and detect recurring advertisements through audio fingerprint matching. This multi-evidence approach reduces ambiguity between acoustically similar music and advertisement segments without end-to-end supervised training. For semantic understanding, we apply LLM reasoning to extract DJ block summaries and music and advertisement metadata from transcripts. Evaluated on 30 days of Korean radio broadcasts, our method achieved F1 scores of 0.91, 0.90, and 0.88 for DJ, music, and advertisement detection, while LLM-based DJ summarization scored 4.42/5 under human evaluation.

Index Terms: automatic speech recognition, radio, audio segmentation, audio acoustic feature, audio fingerprint, speaker diarization, LLM

1. Introduction

A typical long-form radio broadcast program consists of host speech, guest interactions, music tracks, and advertisements. Unlike short audio clips or curated podcasts with a single speaker's speech and explicit metadata, live radio broadcasts are produced in real time and rarely contain fine-grained structural annotations. Consequently, automatic analysis of a one- or two-hour radio broadcast audio stream into structured semantic representations remains a challenging problem.

Conventional audio segmentation approaches mainly rely on local acoustic features and supervised neural classifiers. Although effective for coarse speech–music discrimination, such acoustic-only methods often fail in a long radio program because semantically distinct but acoustically similar segments such as DJ commentary and advertisement speech exist in the same broadcast. Similarly, music tracks and advertisement jingles often exhibit similar acoustic characteristics, making them difficult to differentiate using typical acoustic features. Transcript-based classification alone is also insufficient, as both DJ segments and advertisements share promotional or conversational linguistic patterns.

We argue that long-form radio broadcasts exhibit stable role-driven and temporally recurring structures that can be exploited as global structural priors. A program host, DJ, appears repeatedly throughout the broadcast and serves as a structural anchor over extended time spans. Advertisements typically form clustered segments with short durations and often recur across days, while music tracks follow relatively long temporal lengths. Such role-aware and temporal regularities provide stronger constraints for segmentation than local acoustic evidence alone.

Motivated by this observation, we formulate long-form radio analysis as a role-aware structured prediction problem and propose a hybrid structural modeling framework as shown in Fig. 1. First, we identify the program host, DJ, as a global structural anchor using speaker diarization-based role inference. Second, we separate music and advertisement blocks through acoustic features combined with duration priors. To further improve advertisement detection, we incorporate audio fingerprint matching to identify recurring advertisement segments that appear on multiple days. This multi-evidence modeling reduces ambiguity between acoustically similar music and advertisement segments without requiring task-specific end-to-end acoustic training.

Moreover, to enable semantic understanding beyond structural segmentation, we apply block-conditioned LLM reasoning to extracted transcripts. The LLM summarizes the topic and the summary of a DJ talk block, extracts music metadata from song-related speech, and infers product or company information from advertisement blocks. This modular design bridges structural modeling and semantic extraction, enabling interpretable and scalable long-form broadcast understanding.

We evaluate the proposed framework on 30 days of three major Korean radio broadcast networks, covering different program formats and production styles. Experimental results demonstrate significant improvements in DJ, music, and advertisement block detection. Furthermore, we show that LLMs can perform semantic extraction of a radio program such as DJ talk summaries, music playlists, and advertisement metadata. Our data and source codes are publicly available at this GitHub repository[1].

The main contributions of this work are as follows:

1. We formulate long-form radio analysis as a role-aware structured prediction problem and introduce hybrid multi-evidence structural modeling for DJ, music, and advertisement segmentation.
2. We integrate acoustic features, speaker dynamics, and audio fingerprinting to robustly separate music and recurring advertisement segments in long-form broadcasts.
3. We propose block-conditioned LLM-based semantic extraction for generating structured metadata, including DJ talk summaries, music information, and advertisement entities.
4. We validate the effectiveness of the framework on real-world broadcast data, demonstrating improved structural segmentation and competitive semantic extraction performance.

[1] https://github.com/anonymousresearcher695/interspeech2026

Together, these findings show that role-aware structural priors combined with hybrid evidence modeling and LLM-based reasoning provide a practical and scalable solution for long-form broadcast understanding.

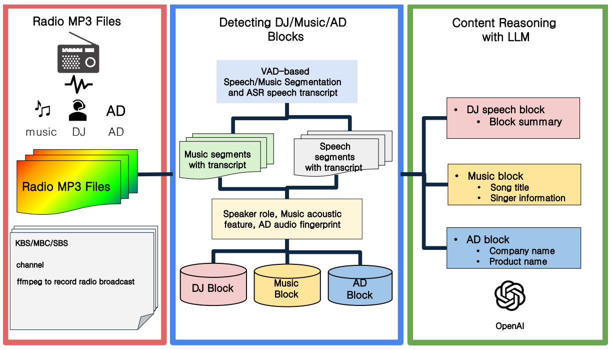


Figure 1: *The overview of the hybrid role-aware structural and semantic extraction model*

2. Related Work

Speech–Music and Audio Segmentation: Early research on audio segmentation focused on speech–music discrimination using acoustic features such as spectral peaks, MFCCs, and energy statistics [5, 4, 1, 2]. Recent work incorporated neural networks and deep acoustic embeddings to improve robustness [9, 3, 6]. Novelty-based segmentation approaches [8] and recent audio-visual segmentation methods [10] further enhanced temporal boundary detection. In the radio domain, Labbé *et al.* [21] proposed a unified framework for segmentation and classification of broadcast audio. However, most existing methods rely primarily on local acoustic evidence and do not explicitly model speaker roles or recurring broadcast structures, which are essential for distinguishing semantically similar segments such as DJ speech and advertisement narration. **Speaker Diarization and Role Inference:** Speaker diarization aims to determine "who spoke when" in multi-speaker recordings. Neural diarization frameworks such as Pyannote [12, 16] and recent end-to-end approaches [13] achieve high accuracy in speaker boundary detection. In broadcast settings, identifying functional roles (e.g., program host versus guest) provides stronger structural cues than speaker segmentation alone. Our approach leverages diarization outputs to infer a structural anchor corresponding to the program host. **Audio Fingerprinting for Recurrent Content Detection:** Audio fingerprinting techniques, such as Panako [14], generate compact acoustic hashes that enable robust matching under noise, compression, and minor time-scale modifications. Fingerprinting has been widely used for music identification and copyright monitoring. In contrast to conventional acoustic classification, fingerprint-based matching is particularly suitable for detecting recurrent and fixed-duration radio advertisements, which often repeat across broadcasts. **LLMs for Audio Understanding:** LLMs including Llama [18] and recent instruction-tuned models such as Qwen [17], have demonstrated strong reasoning capabilities across multimodal and spoken language understanding tasks. Chain-of-thought prompting further enhances structured reasoning in LLMs [19]. Recent work on role-aware reasoning for audio document understanding [20] suggests that incorporating structural context improves document-level interpretation.

3. A Hybrid Role-based Radio Structuring and Reasoning Method

3.1. Problem Formulation

The radio block structuring problem can be defined as follows. Let $X = \{x_t\}_{t=1}^{T}$ denote a long-form radio broadcast audio stream, where x_t represents the acoustic observation at time index t, and T is the total duration of the program. The goal of long-form radio understanding is to transform X into a structured representation consisting of block-level segmentation and semantic metadata.

We define a sequence of non-overlapping blocks

$$B = \{(t_s^k, t_e^k, y_k)\}_{k=1}^{K},$$

where t_s^k and t_e^k denote the start and end times of block k, and $y_k \in \{\mathrm{DJ}, \mathrm{Music}, \mathrm{AD}\}$ represents its structural label corresponding to program host speech, music tracks, and advertisements, respectively. The segmentation task aims to estimate both temporal boundaries and block labels.

Conventional acoustic segmentation models estimate speech or music solely on local acoustic observations. However, long-form radio broadcasts exhibit stable role-driven and temporally recurring structure. We therefore formulate segmentation as a role-aware structured prediction problem. This problem introduces role-aware structural priors that constrain segmentation decisions beyond local acoustic evidence.

Given the structured blocks B, we further extract semantic metadata. A DJ block includes specific topics often with guests, a music block has song title and artist information, and an advertisement block retains product and company entities. The overall objective is to generate a structured semantic timeline that integrates block-level segmentation with content-level understanding.

3.2. Anchoring Speaker Role to find DJ blocks

Given a long-form radio broadcast audio signal X, our goal is to generate a structured semantic timeline consisting of block-level segmentation and metadata extraction. Figure 1 illustrates the overall framework.

The proposed method consists of two main stages: (1) hybrid structural modeling for block segmentation(Alg. 1) and (2) block-conditioned semantic reasoning. The input broadcast is processed with voice activity detection (VAD) to obtain speech and music segments. Automatic speech recognition (ASR) generates transcripts, while speaker diarization assigns speaker labels across the entire program. Based on diarization results, we identify the program host as a global structural anchor using role inference.

Let $\mathcal{S} = \{S_1, S_2, \ldots, S_N\}$ denote the set of detected speakers in the broadcast. To identify the program host as a structural anchor, we compute the following statistics for each speaker S_i:

- T_i: the total cumulative speaking duration,
- O_i: the proportion of speaking duration during the opening interval (e.g., first 5 minutes of the broadcast).

The program host typically exhibits long cumulative duration, frequent temporal recurrence across the broadcast, and prominent presence in the opening segment. We therefore define the DJ, S^* whose total duration for the entire program, T_i and the opening speaking time ratio, O_i are longer than the thresholds.

Algorithm 1 Speaker Role-aware Radio Block Detection

Require: Long-form audio X
Ensure: Block sequence $B = \{(t_s^k, t_e^k, y_k)\}$
Perform VAD to obtain candidate segments $\{(t_s^k, t_e^k)\}$
Perform ASR and speaker diarization
Compute speaker statistics (T_i, O_i)
Determine DJ anchor S^*
for each segment k **do**
  if speaker(k) $= S^*$ **then**
    $y_k \leftarrow$ DJ
  else
    Compute duration d_k
    Compute music acoustic feature M_k
    if AudioFingerprintMatch(k) = True **then**
      $y_k \leftarrow$ AD
    else if $d_k > \theta_d$ **and** $M_k > \theta_v$ **then**
      $y_k \leftarrow$ Music
    else
      $y_k \leftarrow$ AD
    end if
  end if
end for
Merge consecutive segments with identical labels
return B

Segments attributed to S^* are labeled as DJ blocks, while other speech segments are treated as non-DJ speech and further classified into music or advertisement categories in subsequent stages.

3.3. Separation Between Music and Advertisement Blocks

VAD and ASR often generate segmentation inconsistencies for music segments, because both music and AD blocks contain significant levels of vocal speech. To address this, we correct misclassified segments by leveraging music acoustic features. For this purpose, we present a music block filtering algorithm that removes AD candidate segments based on their distinct acoustic characteristics. Music blocks exhibit acoustic characteristics that are distinct from advertisement segments. Specifically, an AD segment usually retains vocal energy equal to or higher than that of background music. In addition, an AD block typically consists of multiple AD segments with short energy dips between consecutive AD segments.

To separate music and AD blocks, we process the entire broadcast audio recording and compute acoustic features, such as vocal and music energy levels, as well as energy drop durations. First, we calculate the *vocal and music energy difference* to filter out DJ speech mixed with background music. Second, we track the *relative energy dip* per 3 seconds, which is the energy drop relative to a moving average, to divide the broadcast audio recording into segments at structural boundaries caused by concatenated AD blocks. Finally, we apply a *minimum valid duration* constraint to identify potential music segments. The segments that successfully pass the music acoustic criteria are labeled as music blocks.

3.4. Advertisement Detection via Cross-Day Audio Fingerprinting

Advertisement blocks often recur across multiple days in radio broadcasts, since many advertisements, campaigns, and program promotions are repeatedly scheduled within the same program. We detect advertisement blocks in the broadcast $X^{(d)}$ by matching audio fingerprints in the previous-day broadcast $X^{(d-1)}$ against $X^{(d)}$ to retrieve acoustically recurring segments. The broadcast is segmented into fixed-length clips using a sliding window with clip length L and step size Δ. Matched clips with scores greater than the threshold θ_s are considered AD segments and are merged into an AD block when the temporal gap between adjacent clips is smaller than θ_g. Unmatched regions between adjacent matched clips are absorbed into the same AD block if the gap is below θ_g. The audio fingerprint-based AD detection procedure is as follows.

1. Split the query broadcast into overlapping clips and retrieve recurring segments using cross-day audio fingerprint matching.
2. Merge temporally adjacent matched clips to form an AD block.

The default parameters used in audio fingerprint-based AD detection are summarized in Table 1.

Table 1: *Default parameters for advertisement block detection.*

Parameter	Value
Query clip length L	20 s
Query step size Δ	3 s
Match score threshold θ_s	100
Match gap threshold θ_g	30 s

3.5. Reasoning Content from Radio Blocks with LLMs

During speech and music segmentation of a radio program, we also generate the transcript for each audio segment as well as the speaker information. For each classified block, we apply an LLM to extract structured metadata from the transcript. For DJ blocks, the model generates a concise summary of the DJ's talk. For music blocks, it identifies the song title and artist from the surrounding context. For advertisement blocks, it infers the advertised product and company name.

For the accuracy of reasoning content for the DJ block, we evaluate the DJ talk summary according to the text summary metrics. In AD block evaluation, we extract the AD metadata such as AD company or product names and compare them with the ground-truth data.

4. Experiments

4.1. Experimental Setup

We collect 30 days of three Korean radio broadcast programs (KBS, MBC, and SBS) spanning three different music genres. For the ground-truth data, we manually labeled each audio segment generated by VAD into DJ, Music, and AD categories, as summarized in Table 2. For audio processing, we employ `Whisper-X`[2][15], which integrates VAD, ASR, and speaker diarization functions, along with `Panako`[7] for audio fingerprinting. For LLM-based reasoning, GPT-4o is adopted to extract structured metadata from each segment.

4.2. Accuracy of DJ, Music, and Advertisement Blocks

We evaluate the accuracy of classifying DJ, music, and AD blocks with precision, recall, and F1 score.

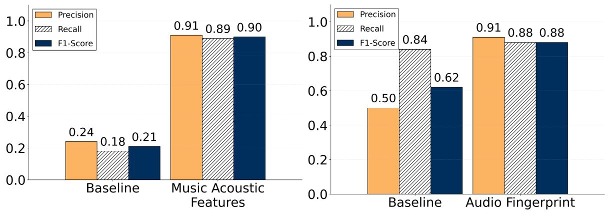
[2]https://github.com/m-bain/whisperX

Table 2: *The radio program dataset*

	Genre	Days	Total Hours	Total Segments
KBS	K-PoP	10	20	8256
MBC	PoP	10	20	8645
SBS	Movie music	10	10	4654
Total		30	50	21555

Table 3: *The accuracy of the proposed DJ block detecting algorithm.*

	Precision	Recall	F1
DJ	0.91	0.92	0.91



(a) *Music* (b) *AD*

Figure 2: *The accuracy comparison of the proposed DJ, Music, and AD block detection method and the baseline.*

- **DJ**: we first report the F1 score of the DJ block detection method in Table 3. Our speaker-based DJ detection algorithm achieved an F1 score of 0.91. By reliably extracting the primary DJ segments first, our model establishes a solid foundation for the overall system. This allows the remaining audio segments to be more efficiently processed and categorized into music and advertisements using acoustic features and audio fingerprinting in subsequent steps.
- **Music**: The baseline of grouping music blocks identifies music segments with long duration (e.g., $\geq$ 100s), which often leads to incorrect segmentation because since vocal-heavy music tends to be fragmented into shorter segments by speech detectors. Our proposed music acoustic feature-based approach in Fig. 2a improves precision to 0.91 and recall to 0.89, boosting F1 score to 0.90.
- **Advertisement**: To evaluate advertisement block detection performance, we compare the baseline with the proposed audio fingerprint-based method. The baseline advertisement detection method groups the audio segments that remain after identifying DJ and music blocks. In Fig. 2b, the baseline method achieves a recall of 0.84 but suffers from a very low precision of 0.50, indicating significant over-detection. In contrast, the proposed audio fingerprint-based approach achieves a precision of 0.91 while maintaining a recall of 0.88. By leveraging cross-day recurrence of advertisement blocks, the method suppresses acoustically ambiguous segments that do not consistently reappear across broadcasts. As a result, the overall F1 score improves from 0.62 to 0.88. Additionally, at the individual advertisement level, the proposed method achieves an F1 score of 0.93.

4.3. Semantic Extraction Results

We evaluated the quality of the reasoned DJ talk summaries manually by a human rater using a 5-point deduction system. A base score of 5 was awarded for a perfectly accurate and comprehensive summary. From this base score, 1 point was deducted for missing essential keywords (e.g., song titles, artist names, or core topics), and 1 point was deducted for any hallucinations or factual errors. The average scores across all three radio broadcast programs are shown in Table 4, with MBC achieving the highest score of 4.54.

Table 4: *Average quality scores for LLM-based DJ block summarization.*

	Days	Blocks	Average Score
KBS	10	100	4.37
MBC	10	124	**4.54**
SBS	10	63	4.48
Total	30	287	4.42

Table 5: *F1 score for LLM-based AD entity extraction (company or product names).*

	AD entities	Precision	Recall	F1
KBS	343	0.77	0.87	**0.82**
MBC	491	0.68	0.80	0.73
SBS	382	0.72	0.88	0.79
Total	1216	0.72	0.85	0.78

For 317 music blocks, we achieved an F1 score of 0.70 in cosine similarity-based extraction of artist and song names. Similarly, advertisement extraction is evaluated via entity-level F1 for company and product names. Table 5 reports the entity extraction performance. The proposed method achieves F1 scores of 0.82, 0.73, and 0.79 on KBS, MBC, and SBS, respectively. A prediction is considered correct if either the company name or the product name matches the ground-truth entity.

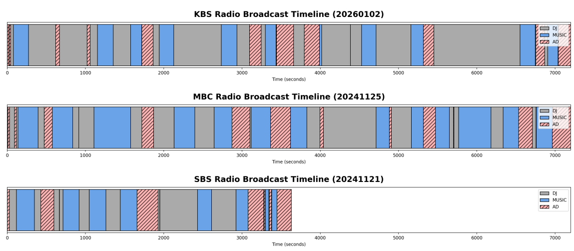


Figure 3: *Broadcast timelines of KBS, MBC, and SBS radio programs with DJ, music, and AD labeling.*

We present timeline plots of three radio programs in Fig. 3. Given DJ, music, and AD blocks by our algorithm, we can understand the different structure of radio programs. Notably, we observe that AD blocks are distributed at approximately 30-minute intervals, appearing consistently at fixed times for each program.

5. Conclusion

This paper demonstrates that structural priors derived from speaker roles and cross-day recurrence provide a practical alternative to large-scale supervised acoustic modeling for long-form broadcast understanding. By combining speaker role inference with music acoustic features and unsupervised audio fingerprint matching, we successfully classify a radio audio stream into DJ, music, and AD blocks. From the DJ, music and AD blocks, we further infer the semantic data with the LLM. We plan to improve the massive data evaluation with years of radio programs and to explore real-time processing.

6. References

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